

Diagnostics of a Multicusp-Assisted Inductively-Coupled Radio-Frequency Plasma Source for Plasma Immersion Ion Implantation

J. Moreno¹, M. Jimenez¹, D. Okerstrom¹, M. P. Bradley^{1*},
and L. Couédel^{1,2†}

¹Department of Physics and Engineering Physics, University of Saskatchewan,
Saskatoon, SK S7N 5E2, Canada

²Aix-Marseille Université, CNRS, PIIM, UMR 7345, 13397 Marseille Cedex 20, France

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Abstract

In this article, we present a detailed characterisation of a multicusp-assisted inductively coupled RF plasma source for plasma immersion ion implantation (PIII). Using laser-induced fluorescence (LIF) and RF-compensated Langmuir probe diagnostics, we measured ion temperature T_i and drift velocity v_z in argon plasmas near an immersed electrode. The multicusp configuration enhances plasma density at low pressure, enabling stable operation down to 0.8 mTorr. Time-averaged measurements show no detectable perturbation near the pulsed electrode, indicating full plasma recovery between high-voltage pulses. LIF-derived potential profiles match Riemann's presheath theory, and ion velocity distributions reveal acceleration consistent with sheath dynamics. These results support the use of LIF for steady-state characterisation of the bulk and presheath regions in PIII systems.

1 Introduction

Modern materials processing demands both precision and efficiency, which in turn require theoretical and computational models validated by empirical data. Among the various surface treatment technologies available, plasma-based processing has emerged as a leading approach due to its versatility and effectiveness across a wide range of applications. These include uniform thin-film deposition [1, 2], multi-tier nanostructure fabrication [3, 4], high-aspect-ratio semiconductor etching [5, 6], and improvements in surface

durability and wear resistance [7, 8].

Historically, techniques such as conventional beam-line ion implantation (CBII) [9, 10], chemical vapour deposition (CVD) [11], and physical vapour deposition (PVD) [12] have been widely used for surface modification. While these methods are still relevant, they present limitations. CBII, for instance, struggles with the uniform treatment of complex geometries and lacks the low-energy ion control needed for ultrashallow junctions in modern microelectronics [13]. Vapour deposition techniques, meanwhile, often suffer from isotropic etching and limited control over ion energy [14].

Plasma Immersion Ion Implantation (PIII) offers a compelling alternative. It enables uniform treatment of irregular surfaces, independent control of ion fluence and impact energy, and a relatively simple system design [15]. PIII has been successfully applied to tribological enhancement [16], thin-film growth [17], etching [18, 19], and the treatment of high-aspect-ratio silicon structures [20]. A more recent application is the use of PIII to study ion bombardment damage to materials intended for fusion reactor plasma-facing components [21]. In PIII, the target is immersed directly in the plasma, and ions are accelerated into the surface by the electric fields that naturally occur in the plasma sheath which surrounds the target. In most applications the target is pulsed with a succession of negative-polarity high-voltage (NPHV) pulses, although in some instances the target can be left floating, or be DC-biased. In the pulsed configuration, the ion fluence is determined by the bulk plasma charge density, while the ion impact energy is governed by the applied voltage [13]. Indeed, the electric field across the sheath between the bulk plasma and a target surface accelerates

*michael.bradley@usask.ca

†lenaic.couedel@usask.ca, lenaic.couedel@cnrs.fr

the ions for deposition and implantation in the PIII system. Applying NPHV pulses to the target modifies ion energy and fluence, enabling the greater control over implantation depth and ion flux. However, accurate modelling of this process is essential for optimising efficiency and yield.

The first model for HV sheath dynamics was proposed by Conrad [22] and later expanded by Lieberman [14]. Lieberman’s model allows the estimation of ion fluence based on steady-state plasma parameters such as density and electron temperature. However, it assumes a stationary and spatially uniform plasma, and a sheath structure that remains consistent across pulses. These assumptions may not hold under realistic operating conditions.

In practice, the NPHV pulse can strongly perturb the plasma [23], even initiating breakdown in the absence of external sources, as seen in pulsed plasma thrusters and high-power impulse magnetron sputtering [24, 25]. This challenges the assumption of steady-state plasma conditions and introduces variability in sheath properties such as potential drop, density, and spatial extent.

Several extensions to Lieberman’s model have been proposed, addressing finite pulse rise times [26], sheath expansion across multiple pulses [27, 28], and collisional effects [29]. Time-dependent sheath behaviour is often explored using Particle-in-Cell and Monte Carlo simulations [28, 30, 31], which show that process efficiency is highly sensitive to pulse repetition rate [32, 33]. The repetition rate must be low enough to allow plasma recovery, yet compatible with thermal constraints imposed by target heating [34]. A sufficiently dense plasma is less affected by the NPHV pulse, improving process reliability.

This article presents the experimental characterisation of a multicusp-assisted, inductively coupled RF plasma source designed for PIII applications. The aim is to provide a stable, low-pressure plasma environment suitable for high-fidelity laser-induced fluorescence (LIF) diagnostics. Section 2 describes the experimental setup and the diagnostic tools. Section 3 presents the results and the discussion.

2 Experimental setup

2.1 Multicusp-Assisted RF Plasma Source

The plasma was generated using an inductively coupled radio-frequency (ICP) source operating at 13.56 MHz, designed by Plasmionique Inc. [35] for the University of Saskatchewan Plasma Physics Laboratory. The system comprises a stainless steel vacuum chamber, RF power

supply with matching network, a pumping system, and a high-voltage pulsing electrode for plasma immersion ion implantation (PIII) studies [36, 23, 37, 21].

The cylindrical chamber measures 29.8 cm in diameter and 46.0 cm in height, and includes eight access ports fitted with ConflatTM flanges for diagnostics such as Langmuir probes and optical viewports. To enhance plasma confinement, permanent ferrite magnets were mounted externally, creating a multicusp magnetic field ranging from approximately 18 G at the chamber edge to 2 G at the centre. This configuration significantly increased ion density (see Fig. 4) and enabled stable plasma operation at low pressures, down to 0.8 mTorr.

A stainless steel electrode was inserted into the plasma for ion implantation experiments. With a surface area of 36.45 cm², it could be grounded, DC-biased, or pulsed using a custom-built Marx-bank modulator. For the experiments described, the electrode was left electrically floating to allow direct ion bombardment. It was mounted along the path of the laser beam and could be manually translated to adjust its depth within the plasma.

High-voltage pulses were applied at 250 Hz with durations of 40 μ s, using a custom Marx-generator type high-voltage modulator [38]. Pulse amplitudes of 500 V and 1 kV were used, corresponding to a 1% duty cycle. Additional measurements were taken with the electrode biased to -30 V DC using a laboratory power supply.

Argon was used as the working gas, with flow rates between 0.5 and 1.0 sccm regulated by an MKS mass flow controller. The vacuum system included a Pfeiffer HiPace 80 turbomolecular pump and a Leybold Turbovac 50, achieving a base pressure of 1 μ Torr prior to plasma ignition. Operating pressures ranged from 0.8 to 2.0 mTorr, chosen to optimise fluorescence signal and maintain high-confinement (H-mode) conditions. Pressures above 3 mTorr led to collisional quenching, while pressures below 0.7 mTorr caused plasma instability and arcing during HV pulsing.

Plasma was ignited using a 600 W RF power supply connected to a water-cooled planar coil antenna. Impedance matching was handled by a Match Pro CPM-1000 network, which used a combination of fixed and variable capacitors to maintain a 50 Ω input impedance. Reflected power was kept below 5 W and the forward RF power was varied between 350 and 525 W to maximise fluorescence signal intensity.

2.2 Langmuir Probe

Plasma parameters such as the electron and ion densities (n_e , n_i), electron temperature (T_e), and plasma potential (V_{pi})—were measured using a rf-compensated

cylindrical Langmuir probe (LP). The probe design follows the RF-compensated model developed by Takashi *et al.* for RF-driven plasmas [39]. The LP uses a tungsten tip, 1.0 mm in diameter and 6.37 mm in length, chosen to ensure sufficient current collection even in low-density regions of the plasma.

To minimise RF interference, the probe is equipped with four inductors tuned to the first two harmonics of the RF coil (13.56 MHz and 27.12 MHz), acting as band-stop filters. These reduce RF-induced fluctuations of the plasma potential, which can otherwise lead to overestimated electron temperatures [40]. Additionally, a copper reference electrode is coupled to the probe tip via an 18 nF capacitor to further suppress RF noise and improve measurement accuracy [41].

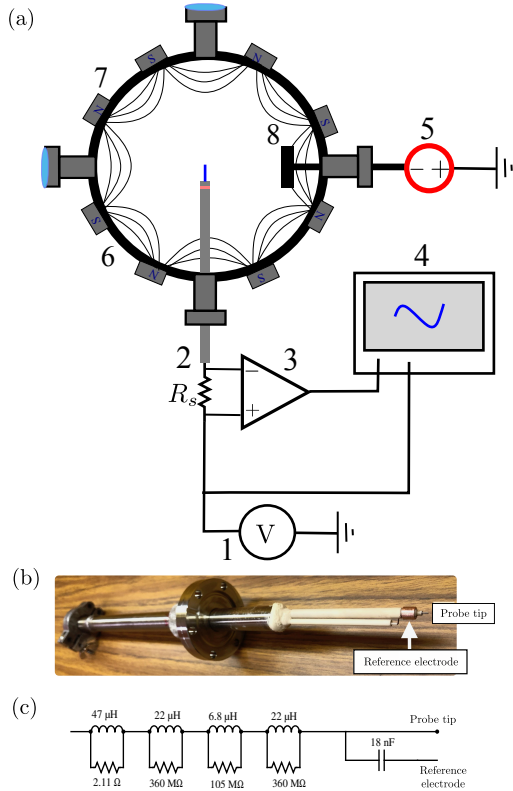


Figure 1: (a) Schematic of the experimental setup for Langmuir probe measurements: 1–Langmuir probe controller adjustable DC power supply; 2–Langmuir probe; 3– operational amplifier; 4–Oscilloscope; 5–High voltage pulser; 6–vacuum vessel; 7–multicusp belt of permanent magnets; 8–Substrate holder. (b) Picture of the Langmuir probe. (c) Schematic of the probe’s internal circuitry, with four inductors connected in series to reduce the effect of the RF fluctuations on the measured signal.

The probe tip was placed along the central axis of the chamber about 15 cm below the rf antenna

(see Fig. 1). Plasma parameters were extracted from the probe IV curves using standard Langmuir analysis [42, 36], which assumes a Maxwellian electron energy distribution. This method was found to be accurate for ICPs in this pressure regime [43], and compared favourably with other techniques such as OML [44], ABR [45], and Druyvesteyn [46], while being computationally efficient for processing large datasets. More details about the probe construction and characteristics can be found in Ref. [37].

2.3 Laser Induced fluorescence system

Figure 2 shows the apparatus used for Laser-Induced Fluorescence (LIF) measurements. This system consists of a tunable diode laser, calibration tools, and signal collection optics. The LIF system is designed to: i) calibrate the laser’s relative frequency axis, and ii) detect and amplify the fluorescence signal.

The light source used is a Toptica DL Pro diode laser, tunable in the range of 659–676 nm, with a linewidth of 600 kHz. The laser wavelength is controlled via adjustments to the grating angle, piezoelectric voltage, diode temperature, and current. A mode-hop-free scan over 20 GHz ensures stable and continuous operation. Laser absorption occurs when the wavelength matches the transition energy of a specific excited ion state in its rest frame. The resulting emission intensity from the newly excited state is recorded as a function of the laser wavelength. Accurate wavelength-resolved intensity measurements enable measurement of the Ion Velocity Distribution Function (IVDF) via the Doppler-shift. The second moment of the IVDF provides a measure of the ion temperature T_i , while the peak position of the absorption profile yields the fluid velocity v_i . In our experiments, the laser was tuned to 668.6138 nm, corresponding to the $3d^4F_{7/2} \rightarrow 4p^4D_{5/2}$ transition of Ar^+ (see Fig. 3) [48].

Calibration is performed using a wavemeter (Bristol 671B, 0.5 pm accuracy) and an iodine reference cell. The I-cell is heated to $\sim 80^\circ\text{C}$ to increase the iodine vapour density, enhancing absorption. Four iodine lines near 668.6138 nm (Fig. 2(b) and (c)) assist in wavelength calibration [47].

Relative frequency calibration is achieved using a home-made Fabry-Perot etalon, which acts as an air-spaced optical cavity. The etalon consists of two confocal mirrors with $\sim 99\%$ reflectivity at 680 nm and a radius of curvature of 44 mm. The mirrors are mounted on translation stages and separated by 4.2 ± 0.1 cm, giving a free spectral range $\Delta\nu_{FSR} = \frac{c}{4\eta d} \sim 1.78$ GHz, where c is the speed of light in vacuum, d is the mirror separation and $\eta \simeq 1$ is the refractive index of air. The output intensity pattern is detected using a silicon photodiode (Thorlabs FD11A), housed in a metal case

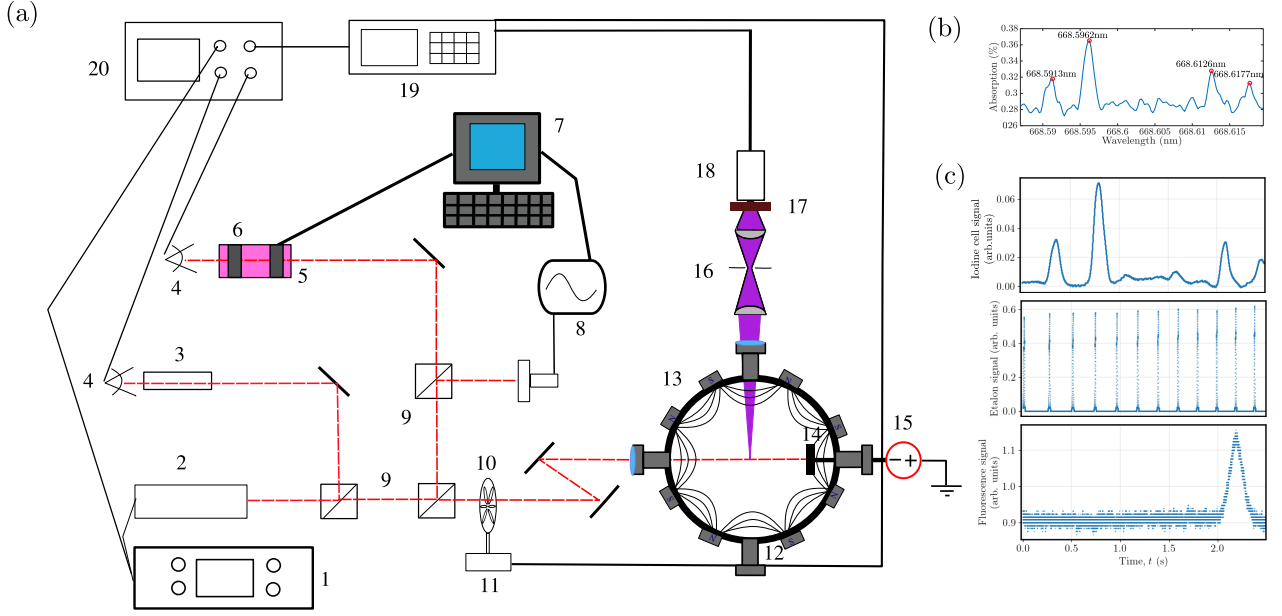


Figure 2: (a) Schematic of the experimental setup for LIF measurements. 1–Laser controller; 2–laser head; 3–Fabry-Perot etalon; 4–photodiode; 5–iodine cell; 6– iodine cell heater; 7–computer; 8–fibre optic head and wavemeter; 9–beamsplitter; 10–chopper blade; 11–chopper controller; 12–plasma chamber; 13–multicusp belt of permanent magnets; 14–pulsing HV electrode; (15) electrode NPHV pulsing system; 16–LIF collection optics; 17–narrow bandpass optical filter; 18–photomultiplier tube; 19–lock-in amplifier; 20–oscilloscope. (b) Theoretical iodine absorption spectrum around the 668.6138 nm argon ion transition [47]. (c) Raw signal from the LIF setup. From top to bottom: Iodine cell absorption spectrum, Fabry-Perrot etalon signal, LIF signal.

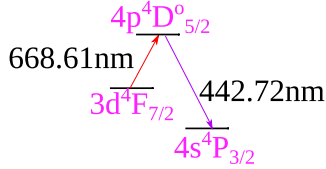


Figure 3: Partial Grotrian diagram for Ar-II transitions.

with a dedicated battery supply to minimise electrical noise (see Fig. 2(c)). The etalon is enclosed in thermal and electrical shielding to ensure stability. The same photodiode setup is used for the iodine cell. Irregularities in the etalon signal indicate mode-hops, prompting further laser tuning.

Detecting fluorescence from optically-pumped plasma ions requires precise filtering and amplification due to the strong background plasma light and electrical noise from the RF generator. The signal is proportional to the density of argon ions in the metastable $3d^4F_{7/2}$ state, which is populated via electron-impact excitation. To enhance ion density without increasing pressure (which causes quenching), multicusp magnetic confinement was introduced using permanent magnets (see Sec. 2.1). Previous studies

showed that Zeeman splitting due to a magnetic field of 1 kG caused a broadening of 3.5 pm on the LIF signal [49]. Assuming the splitting scales with the applied external magnetic field, this would result in a broadening of 0.007 pm in our set-up, which is a negligible affect [50].

The laser beam is modulated by an optical chopper before entering the chamber through anti-reflective coated windows. To minimise reflection from the electrode, the beam is angled slightly off-normal. Fluorescent light is collected using plano-convex lenses and passed through an iris and a narrow bandpass filter centred on the wavelength of the emitted fluorescence photon (442 nm, FWHM 1.0 nm, 45% transmission). The filtered light is detected by a Hamamatsu R3896 photomultiplier tube (PMT), which outputs to a lock-in amplifier (LIA) phase-locked to the chopper frequency. The LIA reduces noise and enhances signal sensitivity. All signals—including from the etalon, iodine cell, laser controller, and LIA—are recorded and averaged by a four-channel oscilloscope to produce the final raw signal [see Fig. 2(c)].

3 Results

3.1 Bulk Plasma Measurements: Power and Pressure Dependence

Langmuir probe measurements conducted at a constant forward rf power of 500 W but varying pressure indicate that the plasma ion density is significantly enhanced in the presence of magnetic cusps. At 0.8 mTorr, the plasma density increases by almost one order of magnitude – from $\sim 6 \times 10^{10} \text{ cm}^{-3}$ at 0.8 mTorr to $\sim 5 \times 10^{11} \text{ cm}^{-3}$ at 3 mTorr (see Fig. 4). This demonstrates the improved confinement of the plasma resulting from the introduction of magnetic cusps. However, the effect diminishes at higher pressures as a result of increased collisionality [51].

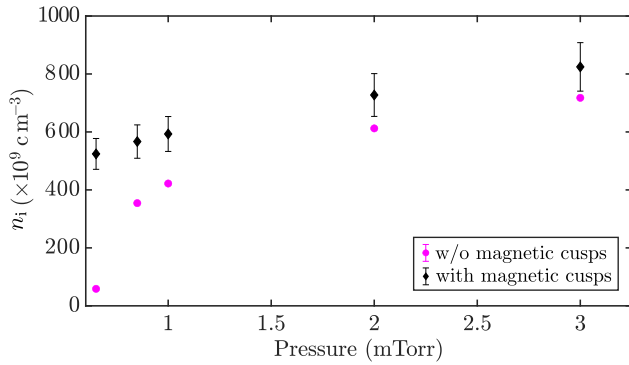


Figure 4: Ion density n_i with (black diamonds) and without (magenta circle) magnetic confinement was measured at various pressures at 500 W input power using a Langmuir probe positioned along the discharge axis. For improved visual clarity, error bars for the measurements without magnetic cusps are omitted; however, they are comparable in magnitude to those associated with the measurements taken in the presence of magnetic cusps.

Then, the first set of LIF measurements combined with Langmuir probe measurements focused on characterising the bulk plasma (i.e. at the center of the vacuum chamber) under varying forward rf power (350–525 W) and neutral argon pressure (0.8–2.0 mTorr). These conditions were chosen to avoid collisional quenching at high pressures (the LIF signal became very weak above 2.0 mTorr) and unstable H-mode of the ICP discharge at low pressures (below 0.8 mTorr). LIF measurements focussed on T_i and v_z while RF-compensated Langmuir probe measurements gave us n_i , n_e , T_e , and V_{pl} (Figure 5).

Langmuir probe measurements must be interpreted carefully. While electron saturation current can indicate n_e , this is reliable only at low densities and pressures, where the electron mean free path is long [40].

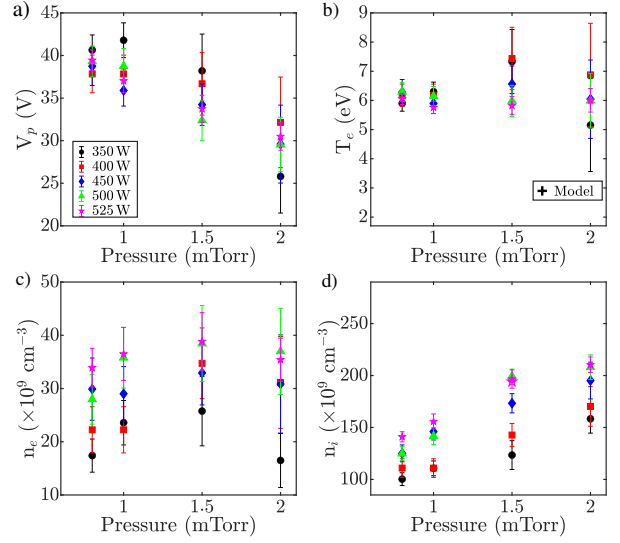


Figure 5: Langmuir probe measurements as a function of RF power and pressure: (a) plasma potential V_{pl} , (b) electron temperature T_e , (c) electron density n_e , and (d) ion density n_i . The electron density n_e is approximately five times lower than the ion density n_i , likely due to probe-induced perturbations that drain the plasma and alter its equilibrium state [40].

At higher densities, the probe can drain the plasma, altering its equilibrium. Therefore, ion current collection is often preferred for estimating density, especially given plasma quasineutrality [40]. Additionally, due to the inherent noise of our data acquisition system, the error bars on the Langmuir probe results were quite large and the obtained values thus served as order-of-magnitude estimates only. The observed trends are (see Fig. 5):

- V_{pl} decreases with pressure and is almost independent of the forward rf power.
- n_i and n_e increase with power and pressure. n_e is approximately five times lower than n_i , likely due to probe-induced perturbations that drain the plasma and alter its equilibrium state [40]. Moreover, due to noise in the IV characteristics, the error bars on n_e are much larger.
- Electron temperature (T_e) ranged from 5 to 7.5 eV and was found to remain relatively constant and seem to depend more on pressure (increasing as pressure decreases) than forward rf power (no trend can be clearly identified).

Our results align well with findings from Langmuir probe diagnostics reported in various studies. Electron temperature (T_e) was observed to depend primarily on pressure, increasing as pressure decreases, while

showing only weak dependence on RF power in certain regimes [52, 53]. In contrast, electron density (n_e) was found to be strongly influenced by both pressure and power, increasing with higher RF power and lower pressure [52]. Across multiple experiments, T_e typically ranged between 2–7 eV [54, 53]. Notably, at low RF power, T_e decreased with increasing power, whereas at high RF power, it increased. This behaviour is attributed to the interplay between neutral gas heating and stepwise ionisation, which modifies the electron energy distribution [55]. Neutral gas dynamics also play a significant role: neutral density is often depleted in the plasma centre due to heating from collisions with charged particles [52]. Furthermore, the plasma heating mechanism transitions from stochastic heating at low pressure to collisional heating at higher pressures. This shift affects how electrons gain energy, particularly in relation to nonlocal electron kinetics and ambipolar potential barriers [56].

The normalised Ion Velocity Distribution Functions (IVDFs) presented in Figure 6(a) exhibit a discernible broadening as the applied power increases, indicative of a higher ion temperature (T_i). This interpretation is corroborated by Figure 6(c), which demonstrates a gradual increase in T_i from approximately 0.70 eV to 0.80 eV over the investigated power range. The dependence of T_i on pressure, also shown in Figure 6(c), is less pronounced; however, there is a tendency for higher pressures to correspond to slightly higher ion temperatures. These findings are broadly consistent with previous investigations, which reported ion temperatures in the range of 0.04–0.08 eV and observed similar trends of increasing T_i with power and, to a lesser extent, with pressure [57, 58, 59]. Notably, Jun *et al.* [59] reported that for pressures exceeding approximately 1.5 mTorr, T_i begins to decrease. The behaviour of T_i at higher pressures remains an open question and merits additional study.

The directed (drift) ion velocity remains slightly negative, approximately -50 m/s, across all investigated powers and pressures [Figure 6(b)]. In an idealised symmetric discharge, the drift velocity at the plasma centre would be expected to be exactly zero. The observed deviation is likely attributable to asymmetries introduced by diagnostic access ports and other structural features of the vacuum vessel, as well as a potential lateral displacement of the LIF interrogation volume from the discharge axis.

3.2 Influence of Proximity to Electrode (Sheath Profile)

This section characterises the ion dynamics in the vicinity of an immersed electrode in our multicusp-assisted inductively coupled argon plasma, with three

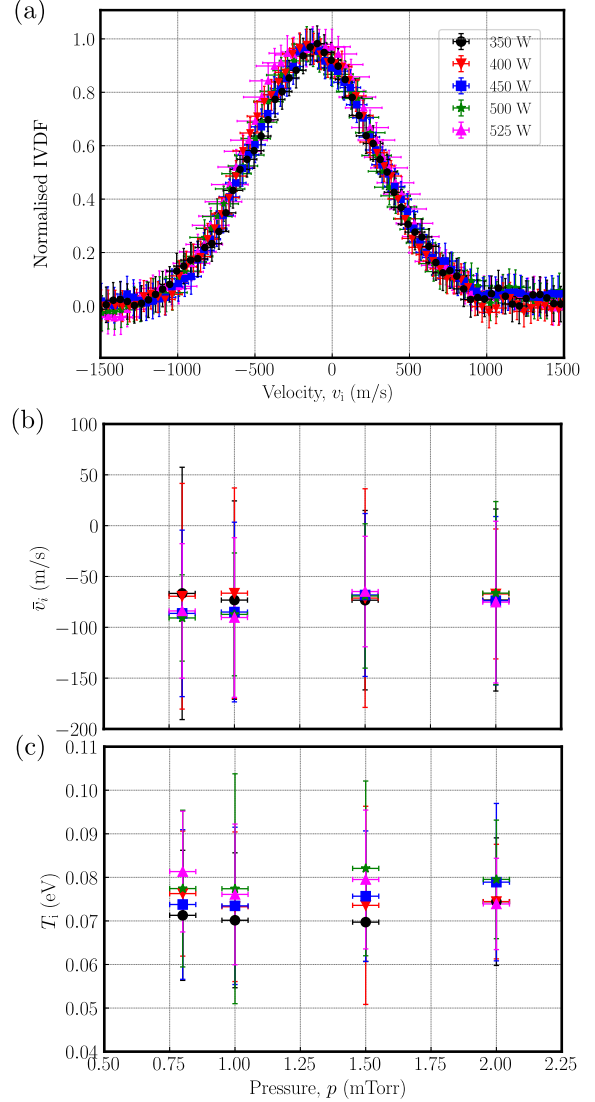


Figure 6: (a) Normalised IVDFs at 0.8 mTorr for various rf powers. (b) Ion drift velocity as a function of pressure for different rf powers. (c) Ion temperature as a function of pressure for different rf powers.

aims: (i) determine whether time-averaged LIF can reveal signatures of the high-voltage (HV) implantation sheath under negative-polarity pulses; (ii) map the directed ion velocity v_z and infer the steady-state presheath/sheath potential profile for grounded and DC-biased conditions; and (iii) examine the evolution of the ion temperature T_i as the measurement volume approaches the electrode.

Experiments were performed at $P_{\text{RF}} = 500$ W and $p = 0.8$ mTorr. The electrode was set to: (a) ground, (b) -30 V DC, and (c) pulsed at -1 kV with 40 μs duration at 250 Hz (1% duty cycle). Spatial scans were executed from $z = 0.1$ to 2.0 cm in front of the electrode (finer steps of 0.1 cm within 0.1 – 0.5 cm).

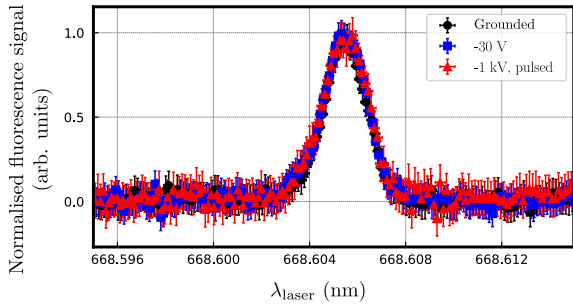


Figure 7: Normalised fluorescence profiles as a function of laser wavelength λ_{laser} measured at a distance of 1 mm from the electrode surface. Three conditions are shown: grounded electrode (black), DC-biased electrode at -30 V (blue), and electrode subject to -1 kV pulsed bias (red). For clarity, the spectral resolution in λ_{laser} has been reduced in the plot.

The time-averaged fluorescence profiles at 1 mm of the electrode for different bias are shown in Fig. 7. No detectable deformation under HV pulsing nor the DC bias can be seen. Consistently, $v_z(z)$ is essentially indistinguishable between grounded, -30 V DC, and pulsed case. Within the averaging limits, this supports the common approximation that bulk parameters between pulses are only weakly perturbed.

Figure 8 presents the normalised experimental IVDFs at different distances d from the grounded electrode. As d decreases, the peak of the distribution shifts towards higher velocities, indicating ion acceleration towards the electrode in the presheath region. The IVDF width appears to decrease progressively down to approximately $d = 4$ mm; beyond this point, the trend becomes less clear. The increase in ion velocity observed close to the target is attributed to ion acceleration by the electric field in the presheath region. The measurements concern metastable argon ions (Ar^{+*} , $3d^4F_{7/2}$), which represent less than 1% of the total ion population [58]. Due to their short mean free paths, these ions are generated locally via electron

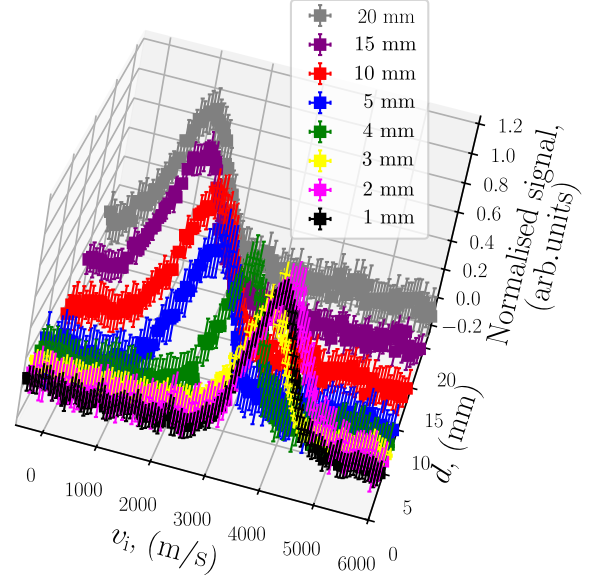


Figure 8: Normalised ion velocity distribution functions (IVDFs) measured at various distances d from a grounded electrode. The profiles illustrate the evolution of the IVDF shape and peak position as the electrode is approached.

impact, rather than being transported from outside the presheath. The absence of a zero-velocity component in the IVDFs confirms that excitation occurs after acceleration. As a result, the measured velocity distribution of metastable ions should accurately reflect that of ground-state argon ions [58].

Equivalent measurements were performed for an electrode biased at -30 V and for an electrode subject to -1 kV pulsed bias (red curves). Each IVDF was fitted with a skewed Gaussian function to quantify asymmetry. From these fits, the mean ion velocity $\bar{v}_i$, the effective ion temperature $T_{i,\text{eff}}$, and the skewness γ_i were extracted as functions of d (Fig. 9). The profiles for all electrode biases are nearly identical, reinforcing the previously-noted assumption that bulk plasma parameters remain largely unperturbed between short-duty-cycle pulses.

The mean ion speed increases monotonically as d decreases, approaching the theoretical Bohm velocity u_B . Based on Langmuir probe measurements at $p = 0.8$ mTorr and $P_{\text{RF}} = 500$ W (Sec. 3.1), the Debye length is estimated as $\lambda_D \approx 40$ – 80 μm , the Bohm velocity as $u_B \approx 3$ – 4 km s^{-1} , and the steady-state sheath thickness as $S_{\text{is}} \approx 0.3$ – 0.6 mm [60]. LIF measurements indicate that ions reach u_B at $d \approx 0.1$ – 0.2 cm, which is slightly larger than the estimated sheath length but remains consistent within the large theoretical uncertainties and the modified conditions introduced by the presence of the electrode.

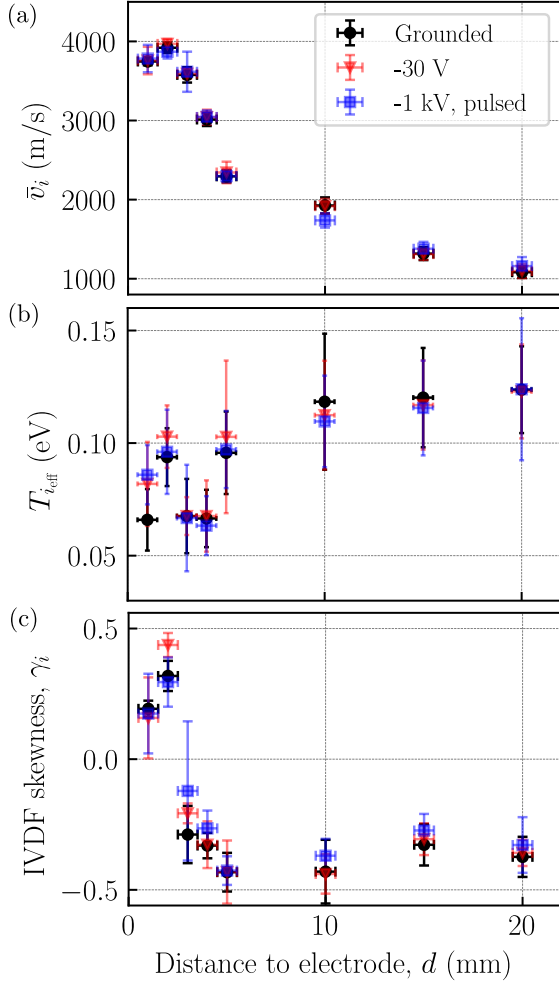


Figure 9: Spatial evolution of IVDF parameters as a function of distance d from the electrode: (a) mean ion velocity $\bar{v}_i$, (b) effective ion temperature $T_{i,\text{eff}}$, and (c) skewness γ_i . Results are shown for grounded, DC-biased, and pulsed-bias conditions.

The fits of the IVDFs indicate that T_i decreases from $d = 20$ to 4 mm and then increases from $d = 4$ to 2 mm (Fig. 9(b)) and seems to decrease again at $d = 1$ mm. This non-monotonic trend contrasts with several earlier reports of monotonically increasing T_i towards a boundary in other devices/conditions [61, 62, 63, 64]. However, the neutral gas temperature profile in the discharge is unknown and it is reported in the literature that the neutral temperature has a maximum in the centre of the discharge and decreases towards the edge [57]. Ion-neutral collision might therefore explain why the ion temperature decreases towards the electrode.

Figure 9(c) presents the evolution of the skewness of the IVDF as a function of the distance d from the electrode. Between $d = 20$ mm and $d = 5$ mm, the skewness is negative and decreases from approximately 0.35

to 0.45, indicating a slight deviation from a Maxwellian distribution. A negative skewness implies an asymmetry in the IVDF, with a higher population of slow ions compared to fast ions. This feature may be attributed to low-velocity ion populations generated by ionisation or charge-exchange processes [65, 66, 67]. As the distance to the electrode decreases further, the skewness increases, reaching zero at $d = 3$ mm (Maxwellian distribution) and becomes slightly positive at shorter distances, where the fast-ion side dominates. It is important to note that the spatial resolution ($\Delta z \sim 0.5$ mm) can introduce artificial broadening of the IVDF due to averaging over the detection volume. Near the target, where ion velocity increases rapidly, this effect may distort the measured IVDF towards higher velocities, leading to an overestimation of both the drift velocity and the ion temperature [68].

4 Discussion

At a pressure of 0.8 mTorr, the neutral density is approximately $n_n \approx 3 \times 10^{19} \text{ m}^{-3}$. Assuming an ion-neutral collision cross section of $\sigma \sim 10^{-18} \text{ m}^2$ [14], the corresponding ion mean free path is $\lambda_{i,\text{MFP}} \sim 3 \text{ cm}$ [14], consistent with previous estimates [69]. This suggests that most LIF measurements are performed well within the spatial extent of the presheath. Using order-of-magnitude inputs from probe diagnostics at 0.8 mTorr and 500 W (Section 2.2), the following plasma parameters are also estimated: Debye length $\lambda_D \approx 50 \mu\text{m}$, ion sound speed $u_B \approx 3.9 \text{ km s}^{-1}$, and ion sheath thickness $S_{\text{is}} \sim 0.3 \text{ mm}$. Using energy conservation, the local potential can be inferred from the axial ion velocity via $\phi_{\text{ps}}(z) = \frac{M_i \bar{v}_i^2}{2e}$. The resulting LIF-derived potential profile can be fitted reasonably well with Riemann's analytical model for a weakly collisional presheath [70, 71]:

$$\phi(z) = T_e \sqrt{\frac{(z - z_0)}{\lambda_{i,\text{MFP}}}} + \phi_0, \quad (1)$$

where z_0 is the position of the sheath-presheath boundary. The fit yields an effective electron temperature of $T_e \approx 3.6 \text{ eV}$ (within a factor of ~ 2 of probe and global model estimates and consistent with typical probe uncertainties) and a sheath presheath boundary at $\sim 1 \text{ mm}$ (in reasonable agreement with our estimate). The measured ion drift velocity $v_z(z)$ reaching the Bohm speed u_B and the agreement of the LIF-derived potential profile $\phi(z)$ with Riemann's model confirm the successful access to the presheath region.

The non-monotonic behaviour of $T_i(z)$ near the electrode likely results from a combination of factors: measurement noise, deviations from Maxwellian IVDFs,

metastable-state kinetics, and spatial variations in neutral gas temperature. The observed IVDF skewness suggests the presence of low-velocity ion populations due to ionisation and charge-exchange processes [61, 62, 63, 64].

Comparisons between grounded, DC-biased, and NPHV-pulsed conditions show that steady-state plasma parameters remain largely unaffected by the NPHV pulses, consistent with the short duty cycle used. Nevertheless, for NPHV pulses of $V_s = -1000$ V, the estimated sheath thickness (~ 2 mm) lies within the LIF collection volume. However, the corresponding Doppler shift (~ 100 GHz) exceeds our laser's tuning range, limiting the direct observation of NPHV sheath dynamics. Time-resolved LIF with extended frequency access is needed to capture transient effects during NPHV pulses.

5 Conclusion

In this study, laser induced fluorescence was used to investigate ion dynamics in the bulk and in the presheath regions of a multicusp-assisted ICP discharge. The system successfully measured T_i and v_z in both bulk plasma and near an immersed electrode.

The key findings are:

- Ion temperature T_i increases with RF power and pressure in the bulk plasma, consistent with prior studies.
- Ion drift velocity v_z remains near zero in the bulk but increases near the electrode, reaching u_B at ~ 1 mm.
- Potential profiles derived from LIF match Riemann's weakly collisional presheath model.
- No time-averaged distortion was observed under NPHV pulses, confirming plasma recovery between pulses.

Future work should focus on improving signal-to-noise ratio and improving the diagnostics capabilities to enable time-resolved LIF and capture the NPHV sheath dynamics. Additional parameter scans and collisional-radiative modelling will help clarify ion temperature behaviour and metastable kinetics. Complementary diagnostics and simulations will support benchmarking of sheath-edge localisation and potential profiles.

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